Manufacturing Architecture

Planning Families, Cost Versions & BOM Generation

Steering Committee — Executive Summary

March 2026

Production Flow — From Raw Wood to Finished Product

Each step is traced in Odoo with lot-level granularity

```
Raw Lot  —>  Process Pack  —>  Autoclave Batch  —>  Dryer Load (= 1 MO)  —>  Finished Lot  —>  RDK Machining
PO receipt      grouping unit      chemical treatment      kiln drying = main MO      FIFO cost layer      subcontractor
```

Planning Family = The Template

Groups products by **species + dimensions + treatment profile**. Only products from the same family can share an autoclave/dryer load.

Family Parameter	Example (KRRS 1")
Stacking hours	4.5 h
Autoclave hours	12 h
Dryer hours	54 h
Scrap % (internal + B-grade)	3.9%
Chemical consumption / m3	676 kg
Typical dryer load volume	22 m3

BOM Auto-Generation

One button — **"Push BoM to Products"** — creates/updates BOMs for every product in the family. No manual BOM editing.

What gets generated per product

- **Wood component:** scrap-adjusted (e.g. 1.0406 m3 raw per 1 m3 FG)
- **Chemical component:** ready-mix consumption per m3
- **Operations:** Stacking, Autoclave, Dryer with family hours

Rule: Change family parameters → re-push. Never edit product BOMs directly.

Cost Version Framework — Budget vs Actual

Structured budget layer enabling gap analysis against actual FIFO costs

Cost Version = One Budget Period

Created monthly or quarterly by the Controller. Contains four building blocks:

Block	Content	Owner
BU Budget	Top-down budget allocation per BU	CFO
WC Rates	EUR/hour per work centre per GL account (personnel, energy, depreciation, maintenance)	Controller
RM Prices	Standard raw material prices	Purchasing
Family Cards	Computed: hours + scrap + WC rates = standard cost/m3	Auto

Lifecycle: **Create → Populate → Review → Lock → Baseline**

Family Cost Card — Example KRRS 1"

Component	EUR/m3	%
Wood (raw material)	291.37	77%
Chemicals (ready-mix)	30.42	8%
Conversion (stacking + autoclave + dryer)	56.75	15%
Total standard cost / m3	**378.54**	**100%**

Gap Analysis

Every product shows **Standard (budget)** vs **Actual (FIFO MO)** with three variance types:

- **Price variance:** (actual price - std price) x actual qty
- **Volume variance:** (actual qty - std qty) x std price
- **Efficiency variance:** (actual hours - std hours) x std rate

Who Does What — Maintenance Workflow

Clear ownership ensures data stays current

Role	Responsibilities	Frequency
Production Manager	• Update family parameters (hours, scrap, capacity) • Push BOMs to products • Assign families to new products	Quarterly or on product change
Controller / CFO	• Create & maintain cost versions • Update WC rates & RM prices • Review cost cards & lock versions	Monthly
System (auto)	• Family cost cards (computed) • Product cost cards (computed) • FIFO layers on MO closing • Gap analysis (std vs actual)	Real-time

Data Architecture Summary

Where is what stored?

Family params	kebony.planning.family
BOMs	mrp.bom (auto-generated)
WC rates	kebony.cost.version.wc.line
RM prices	kebony.cost.version.product.line
Cost cards	kebony.cost.version.family.line (auto)
Actual costs	mrp.production + FIFO layers

Chemical supply chain: Raw chemicals → Mix Production MO → Ready-Mix Tank → Autoclave consumption. Each mix batch creates its own FIFO valuation layer. Standard chemical cost from cost version; variance = actual FIFO vs standard.